

3 -Torque settings

Frame torque settings

Part	Thread/pitch (mm)	Nm ±10% Tolerance	Notes
Headlight fairing			
Screw securing Plexiglas to headlight fairing	M4x0.7	0.2	
Screw securing headlight fairing bracket to headlight support	M6x1	1.5	
Lower screws initial tightening		8	
Upper screws final tightening		8	
Lower screws final tightening			
Screw securing headlight fairing to bracket	M5x0.8	5	
(RH/LH) Rear-view mirror screw	M8x1.25	25	
Side stand			
Sidestand rotation pivot screw nut	M8x1.25	24	
Screw securing sidestand bracket to crankcase	M10x1.5	36	LOCK 2
Stand sensor bolt	M6x1	5	LOCK 2
Front sprocket			
Front sprocket retaining nut	M25x1.5	186 *	
Front sprocket cover screw	M6x1	6	
Lights and reflectors			
Screw securing headlight to support	M5x0.8	5	LOCK 2
Screw securing headlight support to bottom yoke	M6x1	10	
Screw securing front turn indicator to headlight support	M5x0.8	3	LOCK 2
Screw securing headlight unit to support	M5x0.8	3	
Instruments			
Screw securing instrument panel to support	AF5	3	
Instrument panel cover retaining screw	M4x0.7	1.5	
Front Fork			
Steering shaft nut	M35x1	30 *	GREASE B
Screw securing front brake hose guide to bottom yoke	M6x1	8	LOCK 2
Steering limit lock nut	M8x1.25	18	LOCK 2 (apply Loctite to nut thread only)
Screws securing steering head to pivot	M8x1.25	24 *	GREASE B
(RH/LH) bottom yoke to fork screw	M8x1.25	22 *	GREASE B Sequence 1-2-1
(RH/LH) fork bottom end screw	M8x1.25	18 *	GREASE B Sequence 1-2-1

Nut securing clamp to steering head	M10x1.25	45 *	
Screws securing brake/clutch hose guide to steering head	M6x1	10	LOCK 2
Screw securing headlight support plate to steering head	M6x1	10	LOCK 2
Electric system			
Screws securing coil to battery support	M5x0.8	5	
RH/LH handlebar switch screw	M5x0.8	2.5	
Battery cable to terminal fixing screw	M6x1	10	
Nut securing contactor cable to starter motor	M6x1	5	GREASE C
Nut securing battery leads/starter motor to contactor	M6x1	4	
Screw securing battery support to airbox	M5x0.8	5	
Screws securing ECU to airbox	M6x1	10	
Screw securing earth cable to engine	M6x1	10	
Screw securing relay support to battery support	M5x0.8	2	
Screw securing battery support/airbox to frame	M6x1	10	
Oil pressure sensor wiring clamp screw	M6x1	10	LOCK 2
ABS control unit to bracket fixing screw	M6x1	8+/-2	
(Front and rear) ABS sensor retaining screw	M6x1	7	LOCK 2
ABS control unit bracket to airbox fixing screw	M6x1	10	
ABS fuse bracket fixing screw	M5x0.8	7	
Front brake			
Screw securing front brake master cylinder to handlebar	M6x1	10	Sequence 1 (UP)-2-1
(RH/LH) brake calliper coupling screws	M10x1	23	
(RH/LH) front brake calliper screw	M10x1.25	45 *	GREASE B
Preload torque for (RH/LH) front brake calliper screws for settling	M10x1.25	2	GREASE B
Coupling to brake master cylinder fastening screw	M10x1	23	
Special screw retaining couplings onto ABS control unit	M10x1	23	
(RH/LH) brake calliper bleeder retainer	M6x1	4	
Rear swingarm			
Calliper mounting bracket peg fastening pin	M12x1.25	33 *	LOCK 8
Screw fastening lower grommet/chain slider to swinging arm	M5x0.8	5	LOCK 2
Swinging arm pivot nut	M8x1.25	18 *	GREASE B
Chain tensioner eccentric retaining screws	M12x1.25	31 *	GREASE B Sequence 1-2-1
Speed sensor wiring clamp screw	M4x0.7	1	LOCK 2

Screw securing grommet and lower hose guide to swinging arm	M5x0.8	5	
Upper chain slider fixing screw	M5x0.8	5	
Australia chain guard to chain slider retaining screw	AF 5	1.2	Australia version only
Rear brake			
Screw securing brake hose couplings to master cylinder/calliper	M10x1	23	
Screws securing rear brake fluid reservoir to subframe	M6x1	2	
Rear brake calliper bolt	M8x1.25	25 *	GREASE B
Rear brake master cylinder bolt	M6x1	10	LOCK 2
Screw securing rear brake hose guide to swingarm	M5x0.8	5	LOCK 2
Pipe grommet to vertical head fixing screw	M6x1	10	
Pipe grommet to crankcase fixing screw	M6x1	10	
Pipe grommet to crankcase screw retainer	AF3.5	1	
Number Plate Holder			
Screw securing rear turn indicators to protection	M5x0.8	3	LOCK 2
Bolt securing upper to lower number plate holder	M5x0.8	3.5	
Tail light self-tapping screws		3	
Bolt securing X plate to number plate holder	M6x1	5	LOCK 2
Reflector nut	M4x0.7	2	
Number plate holder to protection fixing screw (Australia)	M6x1	5	LOCK 2
Handlebar clutch lever			
Handlebar clamp screw	M8x1.25	25 *	GREASE B Sequence 1-2-3-4-3
Clutch bleeder coupling retainer	M6x1	4	
Clutch hose coupling retainer	M10x1	23	
Clutch slave cylinder screw	M6x1	10	Sequence 1-2-3-1
Throttle twistgrip fixing screw	M6x1	6	Sequence 1 (UP)-2-1
Clutch master cylinder bolt	M6x1	10	Sequence 1 (UP)-2-1
Throttle twistgrip housing screw	M4x0.7	1.8	
Throttle control cable retainer fixing screw	M10x1.25	4	
Throttle control cable adjuster retainer lock nut fastener	M6x1	2.5	
Rear suspension			
Shock absorber clevis fixing screw	M10x1.25	42 *	GREASE B
Screw securing shock absorber to swing arm	M10x1.5	42 *	GREASE B

Front Mudguard Assembly			
(RH/LH) front mudguard screw	M6x1	5	LOCK 2
Front pipe grommet to mudguard fixing screw	AF 3.5	0.4	
Special screw fastening the front footrests to swingarm pivot	M12x1.25	55 *	GREASE B
Front Mudguard - Brake-Side			
Pivot fastening brake lever to right-hand footrest bracket	M8x1.25	24	LOCK 2
Rear stop switch special screw	M6x1	5	
Master cylinder pushrod retaining nut	M6x1	7.5	
Screw retaining right-hand footrests mount to engine	M10x1.5	36	
Brake lever adjuster nut	M6x1	2.3	
Front Mudguard - Gearbox-Side			
Joint to gearchange rod right-hand nut fastening	M6x1	5	LOCK 2
Screw retaining left-hand footrests mount to engine	M10x1.5	36	
Joint to gear change rod left-hand nut fastening	M6x1	5	LOCK 2
Linkage and gearchange lever tie-rod junction fixing screw	M6x1	10	
Screw securing gearchange lever to selector claw	M6x1	10	LOCK 1
Gearchange lever pedal pivot fastener	M8x1.25	24	LOCK 2
Rear Mudguard Assembly			
Rear mudguard special bolt	M5x0.8	5	LOCK 2
Rear mudguard to brake pipe grommet special screw	M5x0.8	5	LOCK 2
Front wheel			
(RH/LH) brake disc screw	M8x1.25	25 *	LOCK 2
Front wheel nut	M25x1.25	63 *	GREASE B
Rear wheel			
Brake disc screw	M8x1.25	25 *	LOCK 2
Rear sprocket nut	M10x1.25	48 *	LOCK 2
Rear wheel shaft nut (left-hand; rear sprocket side)	M33x1.5	156 *	GREASE B
Rear wheel shaft nut (right-hand; rim side)	M38x1.5	176 *	GREASE B
Bolts securing driving pins	M6x1	11	LOCK 2
Phonic wheel fixing screw	M5x0.8	5	
Fuel tank			
Screw securing fuel tank support brackets to fuel tank	M6x1	8	
Screw securing fuel pipe guide to fuel tank	M5x0.8	5	

Flange to fuel tank retaining screw	M5x0.8	5	LOCK 2
Tank filler cap screw	M5x0.8	3	
Special screw (double male thread) fastening flange to tank	M5x0.8	5	LOCK 2
Flange cover flanged nut	M5x0.8	3	
Screw retaining tank side/central covers and central cover to tank cap	M5x0.8	2	LOCK 1
Screw securing front central clip cover to side covers	M5x0.8	2	LOCK 1
Screws securing tank side covers to fuel tank (lower side)	M5x0.8	2	LOCK 1
Screw securing cover to fuel tank (hidden)	M5x0.8	2	LOCK 1
Exhaust			
Screw retaining exhaust sec. to tank/exhaust bracket (left-hand side)	M6x1	5	
(Horizontal/vertical) exhaust pipe clamp fastener	M8x1.25	11	
Screws securing silencers to subframe	M6x1	10	
Nut securing exhaust pipe flange to horizontal/vertical cylinder head	M8x1.25	6	
Nut 1 initial tightening		24	
Nut 2 final tightening		24	
Nut 1 final tightening			
1/8" gas stainless steel plugs to exhaust pipes	M10x1.25	25	
(RH/LH) rear screw retaining heat guard to silencers	M6x1	10	LOCK 5
Screw securing silencer to silencer support on subframe	M8x1.25	22	
Lambda sensor fastener (horizontal/vertical)		45	
(RH/LH) front screw retaining heat guard to silencers	M6x1	8	LOCK 2
Screw securing heat shield to horizontal exhaust	M6x1	8	LOCK 2
Exhaust valve cover screw	M6x1	10	
Screw retaining exhaust sec. to tank/exhaust mounting bracket (right-hand side)	M6x1	5	
Screw retaining exhaust sec. to right-hand footrest bracket	M6x1	5	
Seat			
Seat cover special bolt	M6x1	5	
Frame			
Flanged nut securing frame/subframe to engine	M12x1.25	60 *	Apply GREASE B (to the thread and the underside of the nut)

Special screw securing frame to subframe	M12x1.25	60 *	GREASE B
Special ring nut fastening to clearance compensator dowel	M25x1	100 *	GREASE B
Clearance compensator dowel fastening		0.6	GREASE B (to thread only; keep mating surface free of grease)
Screw securing LH side fuel tank pin to frame	M6x1	10	
Ignition switch cover retaining blind nut fastening	M6x1	7	
Ignition switch special screw	M6x1	Shear screw head	LOCK 2
Screw securing horn bracket to frame	M6x1	10	LOCK 2
Screws securing fuel tank support plate to subframe	M6x1	10	
Screws securing subframe joining plate	M8x1.25	22	
Bolts securing subframe joining plate to rear tile assembly	M6x1	8	
Screw securing fuel tank to bracket on rear subframe	M8x1.25	10	LOCK 1
Seat lock nut	M22x1.5	3	
Screw securing under seat storage compartment	M5x0.8	5	
Special screw securing latch to subframe jointing bracket	M6x1	10	
Screw securing wiring clips to RH rear subframe	M4x0.7	2	
Nut securing rear shock absorber clevis to left-hand side subframe	M10x1.25	42 *	GREASE B
Exhaust valve actuator screws	M6x1	10	
Oil cooler			
Radiator fixing screw	M6x1	10	
Oil hose grommet retaining screw	M6x1	10	
Oil cooler nipple to cooler fastener	M14x1.5	23	Counterhold oil cooler with LOCK 1
Oil pipe union to cooler nipple fastener with O-ring (TWIN SEAL)	M16x1.5	18 *	Engine oil
Oil pipe union to casing nipple fastener with O-ring (TWIN SEAL)	M16x1.5	25 *	Engine oil
Thermistor fixing		10	LOCK 4
Air intake oil breather			
Screws securing filter cover to airbox	M5x0.8	5	
Air filter retaining nut	M6x1	6	
Screws securing airbox to frame	M6x1	10	
Clamp securing oil breather valve hose to blow-by valve		1.5	

Clamp securing blow-by valve hose to airbox		1.5	
Bolt securing coil to airbox	M5x0.8	5	
Clamp securing hoses to airbox		1.1	
Screw securing cable guide to LH side of crankcase	M5x0.8	5	
Fuel system			
Temperature sensor fixing to intake hose	M3x0.5	0.4	LOCK 6
Clamp securing hose to throttle body		1.5	
Clamp securing throttle body to intake manifold		1.5	
Stepper tube tie		10	
EVAP system canister filter			
Canister fixing to LH air cowling	M6x1	8	California version only
Fairings			
Lower special screw fastening right and left-hand side deflector to frame	M5x0.8	5	
Special screw fastening right and left-hand side deflector to central conveyor	M5x0.8	5	
Upper special screw fastening right and left-hand side deflector to frame	M5x0.8	5	
Screw securing voltage regulator to RH air cowling	M6x1	8	

* dynamic safety-critical point; tightening torque must be within Nm $\pm 5\%$.



Note

For product specifications and symbols, refer to paragraph "[Product specifications](#)" (Sect. A 2).

Engine torque settings

Part	Thread (mm)	Nm	Min.	Max.	Notes
Camshaft timing belt pulley nut	M15x1	71	64	78	GREASE A
Cylinder head nuts	M10x1.5				Apply GREASE C to face under cylinder head and stud threads
Stage 1		15	13	17	
Stage 2		30	28	32	
Stage 3		48	46	50	
Intake flange stud	M8x1.25	15	13	17	LOCK 2 or THREE BOND 1324
Intake manifold nut	M8x1.25	23	20	26	
Exhaust flange stud	M8x1.25	15	13	17	LOCK 2 or THREE BOND 1324
Timing hole cover	M8x10	10	9	11	
Vertical cylinder oil temperature sensor screw	M8 x1.25	9	8.5	9.5	
Cylinder head service screw (aluminium)	M6x1	3	2	3	LOCK 5
Intake/exhaust valve cover screws	M6x1	10	9	11	
Valve cover screws	M6x1	10	9	11	LOCK 2
Timing belt cover dome head screw	M6x1	10	9	11	LOCK 2 or THREE BOND 1324
Timing belt cover screw	M6x1	10	9	11	LOCK 2 or THREE BOND 1324

Pulley flange retaining screw	M6	10	9	11	
Canister plug screw	M6x1	5	4	6	
Canister nipple (USA version)	M6x1	5	4	6	
MAP sensor coupling screw	M5x0.8	5	4	6	
Pick-up filter housing plug	M32x1.5	42	38	46	
Oil pick-up filter	M22x1.5	42	38	46	
Oil cartridge nipple	M16x1.5	42	38	46	LOCK 2 or THREE BOND 1324
Oil cartridge	M16x1.5	11	10	12	Engine oil on gasket
Oil pump intake service hole plug	M15x1	25	22	28	LOCK 5
Oil cooler nipple	M14x1.5	32	29	35	LOCK 5
Oil drain plug with magnet	M12x1.5	20	18	22	TB1215
Blow-by hole service plug - Tecopress new	M12x1.5	15	16	17	LOCK 5
Main oil way plug	M10x1.5	15	13	17	LOCK 5
Clutch cover inner oilway screw	M10x1.5	15	13	17	LOCK 5
Neutral sensor	M10x1.25	10	9	11	
Cylinder barrel/head stud	M10x1.5	30	28	32	LOCK 2 or THREE BOND 1324
Taper blanking plug for service hole	M10x1	20	18	22	LOCK 5
Oil pump adapter bolt	M10x1				LOCK 5
Crankcase screw	M8x1.25				Apply GREASE B to thread
Initial tightening		19	17	21	
Final tightening		25	22	28	
Gear stop lever bolt	M8x1.25	18	16	20	LOCK 2 or THREE BOND 1324
Crankcase screw	M6x1	10	9	11	
Crankcase screw	M6x1	10	9	11	
Blow-by valve screw	M6x1	10	9	11	
Radial pick-up screw	M6x1	10	9	11	
Clutch side primary bearing retaining plate screw	M6x1	10	9	11	LOCK 2 or THREE BOND 1324
Chain side secondary bearing plate retaining bolt	M6x1	10	9	11	LOCK 2 or THREE BOND 1324
Idler gear shaft retaining screw	M6x1	10	9	11	LOCK2 or THREE BOND 1324
Adler clutch oil calibration screw	M6	6	5	7	
Under-piston oil jet retaining screw	M5x0.8	8	7	9	LOCK2 or THREE BOND 1324
Starter motor mounting screw	M6x1	10	9	11	LOCK 2
Outer starter motor mounting screw	M6x1	10	9	11	LOCK 2
Starter motor rear screw	M6x1	10	9	11	LOCK 2
Flywheel nut	M24x1	270	256	284	LOCK 5
Primary drive gear nut	M22x1	190	171	209	GREASE B
Steel/aluminium crankshaft blanking cap	M20x1	15	13.5	16.5	LOCK 5
Connecting rod screw	M10x1				Apply GREASE B to thread
Tighten at 50 rpm		35			
2 second pause, then back off through 360° at 30 rpm		20			
Tighten at 30 rpm		35			
Tighten at 10 rpm					
Tighten through 65° +/- 1° at 14 rpm		70 to			
Torque check		103			
Crankshaft grub screw	M8x1.25	13	11	15	LOCK 5
Drilled crankshaft grub screw	M8x1.25	13	11	15	LOCK 5
Selector drum locating screw	M16x1.5	30	27	33	

Selector claw screw	M8x1.25	36	34	38	LOCK 2 or THREE BOND 1324
Selector claw locator nut	M6x1	10	9	11	
Selector claw screw	M6x1	16	15	17	LOCK 2 or THREE BOND 1324
Timing belt driveshaft pulley retaining nut	M15x1	71	64	78	GREASE A
Timing belt driveshaft gear nut	M14x1	55	50	60	GREASE A
Tensioner pulley retaining screw	M8x1.25	26	23	29	
Idler pulley retaining screw	M8x1	20	18	22	LOCK 2 or THREE BOND 1324
Plastic cover screw	M6x1	10	9	11	
Carbon fibre cover screw	M6x1	6	5.5	6.5	LOCK 2
Oil filler plug	M22x1.5	-	-	-	Tighten as far as it will go
Soundproofing gasket fixing screw	M14x1.5	24	22	26	LOCK5
Pick-up sensor inspection screw	M12x1	15	13	17	LOCK 2 or THREE BOND 1324
Clutch cover oilway blanking plug	M10x1	20	18	22	LOCK 5
Oil pressure sensor	M10x1	19	17	21	
Clutch cover screw	M6x1	10	9	11	
Dry clutch cover screw	M6x1	10	9	11	
Generator cover screw	M6x1	13.5	12.5	14.5	
Timing inspection cover screw	M6x1	5	4.5	5.5	
Generator stator retaining screw	M6x1	10	9	11	LOCK 2 or THREE BOND 1324
Stator wiring clamp bracket screw	M5x0.8	5	4.5	5.5	
Generator cover inspection plug	M15x1	20	18	22	THREE BOND 1215
Dry clutch centre nut	M25x1.5	250	236	264	LOCK 5
Dry clutch drum screw	M8x1.25	35	33	37	LOCK 4
Dry clutch spring screw	M5x0.8	5	4.5	5.5	
Spark plug	M12x1.25	20	18	22	
Alternator rotor/flywheel screw	M6x1	13	11	15	LOCK 5
By-pass pump cap	M15x1	17	15	19	LOCK 5
Oil pump retaining screw	M8x1.25	26	23	29	
Oil pump retaining screw	M6x1	10	9	11	
Oil pump assembly screw	M6x1	10	9	11	



Note

For product specifications and symbols, refer to paragraph "[Product specifications](#)" (Sect. A 2).